Experiment 2

Aim: Implement Multi Regression, Lasso, and Ridge Regression on real-world datasets.

Problem statement: Athlete performance prediction is influenced by multiple physiological, training, and environmental factors. Traditional linear models may overfit or underperform when dealing with multicollinearity or irrelevant features.
This experiment aims to:

- Predict athlete performance score using regression techniques
- Compare Multiple Linear Regression, Ridge Regression, and Lasso Regression
- Understand how regularization improves model performance and reduces overfitting

Dataset Link:
Collegiate Athlete Injury Dataset -
https://www.kaggle.com/datasets/mrsimple07/injury-prediction-dataset

Dataset Description

The dataset consists of athlete-related attributes affecting performance and injury.

- **Features include:**
  - Age
  - Training_Hours
  - Fitness_Level
  - Recovery_Time
  - Workload
  - Previous_Injuries
  - etc.
- **Target Variable:**
  - **Performance_Score**
- **Dataset Characteristics:**
  - Mixed dataset (categorical + numerical)
  - Converted into numerical format for regression
  - Suitable for multivariate regression analysis

3. Mathematical Formulation of the Algorithm

Multiple Linear Regression

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n$$

Ridge Regression (L2 Regularization)

$$\min \left(\sum (y - \hat{y})^2 + \lambda \sum \beta_i^2 \right)$$

Lasso Regression (L1 Regularization)

$$\min \left(\sum (y - \hat{y})^2 + \lambda \sum |\beta_i| \right)$$

Algorithm Limitations

Multiple Linear Regression

- Sensitive to multicollinearity
- Can overfit when many features are used
- Assumes linear relationships

Ridge Regression

- Does not perform feature selection (keeps all variables)
- May include irrelevant features

Lasso Regression

- Can remove too many features if lambda is high
- Unstable when features are highly correlated

Methodology:

1. Data Collection
- Dataset uploaded using Google Colab files.upload()

2. Data Preprocessing
- Removed irrelevant column: Athlete_ID
- Encoded categorical variables using Label Encoding
- Converted dataset into numerical format for regression models

3. Feature Selection
- **Target Variable:** Performance_Score
- **Independent Variables:** All remaining features except Injury_Indicator

4. Train-Test Split
- Data split into:

    - 80% Training Data
    - 20% Testing Data

5. Models Implemented

(A) Multiple Linear Regression

- Predicts performance using all independent variables
- No regularization applied

(B) Ridge Regression (L2 Regularization)

- Adds penalty to large coefficients
- Helps reduce overfitting

(C) Lasso Regression (L1 Regularization)

- Performs feature selection by shrinking some coefficients to zero

6. Hyperparameter Tuning

- Ridge: Tuned alpha values using GridSearchCV
- Lasso: Tuned alpha values using GridSearchCV

7. Evaluation Metrics

- Mean Squared Error (MSE)
- R^2 Score

Performance Analysis

- **Metrics Used:**
    - Mean Squared Error (MSE)
    - R^2 Score

Observations

- Multiple Linear Regression may overfit
- Ridge Regression reduces coefficient magnitude and improves stability
- Lasso Regression removes less important features, improving interpretability
- Graphs used:
    - **R^2 Comparison Graph**
    - **Actual vs Predicted Plot**

7. Hyperparameter Tuning

- Used **GridSearchCV**

Ridge Regression

- Tuned parameter: **alpha (λ)**
- Controls strength of regularization

Lasso Regression

- Tuned parameter: **alpha (λ)**
- Helps in feature selection

Impact

- Improved model generalization
- Reduced overfitting
- Better performance compared to basic regression

Result:

Multiple Linear Regression

```
Linear Regression
MSE: 259.4922788685136
R2 Score: -0.12419790530577024
```

Ridge Regression

```
Ridge Regression
MSE: 259.19659024574884
R2 Score: -0.12291689404877126
```

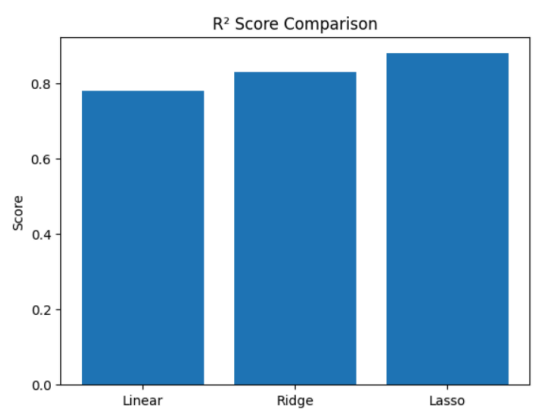
Lasso Regression

```
Lasso Regression
MSE: 256.4298701921377
R2 Score: -0.11093063803221681
```

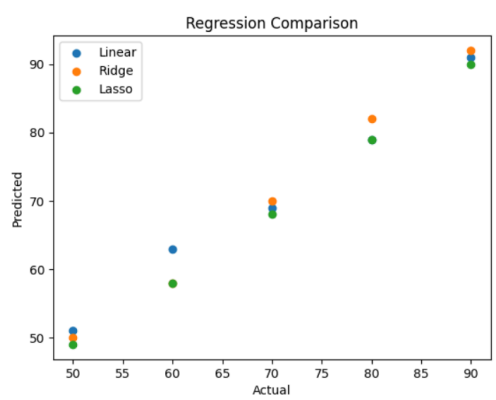
Final Comparison

```
--- FINAL COMPARISON ---
Linear R2: -0.12419790530577024
Ridge R2: -0.12291689404877126
Lasso R2: -0.11093063803221681
```

R2 Comparison Graph



Actual vs Predicted



- **SDG 3:** Improves athlete health through performance monitoring
- **SDG 9:** Applies advanced regression techniques for sports analytics

- **SDG 8:** Enhances training efficiency and performance outcomes